



NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

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Workout Progress Tracker

Stage 2: User and task analysis



Authors:

75346 Matej Palider
75267 Kacper Kuźnik
75160 Dawid Bogacz
75231 Mikołaj Nowacki

Labclass N° P2

Group N° 12

Professor:
Teresa Romão

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Problem

Most workout apps focus on pre-made routines with fixed numbers of sets and reps, but they don't effectively track progress during each session. Users often have to log results manually or remember them, which creates distractions and reduces motivation. A system that records every set and rep in real time and displays progress clearly during workouts can make training more engaging, organized, and effective.

User analysis

The target users of the workout tracking app can be divided into two main groups:

1. Individual Exercisers

- Anyone who wants to work out and track their progress, ranging from young beginners to elderly users.
- Includes people doing weightlifting, calisthenics, yoga, stretching, or rehabilitation exercises.

2. Trainers and Coaches

- Fitness professionals who supervise clients.
- Can be personal trainers, physiotherapists, or sports coaches.

Task Analysis

Task 1: Setup & Planning

Objective: The user wants to create and organize a workout plan so they can follow a structured routine.

Precondition: User has a registered account, user is logged in into system

Exceptions: incomplete client data, database access failure

1. **Workout type selection** - weights, calisthenics, yoga, stretching, rehab, cardio...
2. **Add exercises** - Pick from a predefined library of exercises, add custom/manual exercise.
3. **Define sets/reps/hold**
4. **Set rest time** - Define rest periods between sets and exercises.
5. **Name workout plans** - Add custom titles, add optional notes.
6. **Categorize plans** - Group workouts into categories (legs, arms, yoga, full-body).
7. **Exercise recommendations** - Show suggested exercises based on user history.

Frequency: Whenever a user creates a new workout plan (can be once a month, can be every 6 months)

Task 2: Performing the Workout

Objective: The user wants to track their workout in real time so they can stay focused without distractions.

Precondition: User has a registered account, user is logged in into system, user has created/assigned workout plan, app has necessary permission(for measuring heart rate)

Exceptions: incomplete client data, invalid workout, sensor error

1. **Start workout session** - Select a saved plan or create a quick workout on the spot, preview the workout flow before starting.
2. **Track each set** - Tap to log completed reps/hold, input weight for each set (auto-fill from last workout)
3. **Auto-start rest timer** - Trigger countdown after set completion.
4. **Log heart rate & other metrics** - Manually input heart rate/Sync with wearable device (smartwatch, HR strap).
5. **Provide live feedback** - Show reps completed and sets left, display rest countdown timer, highlight progress toward workout completion.
6. **Workout modification during session** - Skip or replace an exercise, add an extra exercise, change rest time or number of sets.
7. **End workout session** - Save session results

Frequency: On every workout (approx: 1-7 times per week)

Task 3: Reviewing Progress

Objective: The user wants to review their past workouts to track improvements and identify weaknesses.

Precondition: User has a registered account, user is logged in into system, user have some workout history - otherwise the graf will be empty

Exceptions: database access failure, corrupted data

1. **View session summary** - Show total sets/reps completed, display weights used and duration, highlight heart rate (average, peak).
2. **Track long-term progress** - Provide workout history by day, week, month. Highlight streaks (number of consecutive workout days/weeks).
3. **Compare performance** - Compare current vs. past session results. Show improvements.
4. **Identify improvements** - Highlight PRs (personal records). Offer suggestions for adjustments.
5. **Generate reports** - Provide weekly/monthly summaries. Export data (CSV, PDF) for trainers or personal use.

Frequency: After every workout and whenever reviewing progress (1-7 times per week or more)

Task 4: Assign Workouts to Client (Trainer Role)

Objective: Allow trainers to create, assign, and monitor workouts for their clients.

Precondition: User has a registered account with trainer privileges, user is logged in into system, user has at least one active client, stable internet connection, user has created one or more workout plans

Exceptions: connection error, duplicate assignment

1. **Assign plan to client** - Assign single or multiple plans. Set schedule (days of the week).
2. **Monitor client activity** - View completed sessions. Check client progress and performance stats.
3. **Provide feedback** - Leave notes, tips, or instructions.
4. **Manage multiple clients** - Dashboard for switching between clients. Group assignments (e.g., same plan for multiple clients).

Frequency: After every workout, whenever meeting a new client, or when the client progresses in difficulty.

Task 5: User Management & Personalization

Objective: Provide secure and personalized user access to the system.

Precondition: User has a registered account, stable internet connection

Exceptions: connection error, invalid credentials

1. **Sign in**
2. **Profile setup**
3. **Account management** - Edit profile and preferences. Manage notifications and reminders. Set privacy options (trainer access, public/private progress).

Frequency: Not frequent. Depends on user preference to manage the account. Sign in will be automatic.

Scenarios

Scenario 1: Exerciser Completing a Workout

Actor: Individual user (fitness enthusiast).

Goal: Perform a workout and record progress.

Context: User is at the gym and wants to focus on training without distractions.

The user logs into the app and sees today's scheduled workout.

They choose the plan they want to follow and begin the session.

As they train, the system helps them record completed sets, repetitions, and weights used, filling in defaults from previous sessions where possible.

Rest times are tracked automatically, and the system provides ongoing feedback about progress through the workout.

If the user decides to make changes (e.g., skip an exercise or add an extra set), the system allows flexible adjustments during the session.

Once finished, the system generates a summary of the workout, including sets completed, weights used, heart rate, and overall duration.

Scenario 2: Trainer Assigning Workouts to Clients

Actor: Personal trainer.

Goal: Create, assign, and monitor workouts for clients.

Context: Trainer is preparing training plans for the upcoming week.

The trainer accesses the app with their professional account and views their client list. They create a new structured workout, adding relevant exercises, sets, repetitions, and rest times.

The workout is given a descriptive name and categorized for easy reference.

The trainer assigns the plan to specific clients.

The system provides a dashboard overview of multiple clients' activities.

The trainer assigns the plan to specific clients.

Scenario 3: User Reviewing Progress

Actor: Returning user.

Goal: Understand progress and adjust future training.

Context: User wants to see how their performance has evolved over time.

After logging in, the user navigates to their workout history.

They browse past sessions and review details such as sets, repetitions, weights, and heart rate trends.

The system highlights progress, such as personal records, increased loads, or improved endurance.

Graphs and summaries show long-term development, streaks of consistent training, and comparisons to earlier periods.

Interviews

Interview 1 - Yoga

User background:

Intermediate yoga enthusiast who practices daily at home and sometimes in studios. Uses apps mostly for pose guidance and progress tracking.

Feedback:

- Likes the idea of tracking sessions but feels yoga isn't about "sets/reps"; prefers time-based and flow-based tracking (e.g., how long a pose was held).
- Would love pose history; how long specific poses were held over time, and improvement in flexibility or balance.
- Feature to consider: Mindfulness tracking (e.g., meditation time, mood before/after session).

Interview 2 - Calisthenics

User background:

Advanced calisthenics practitioner, is able to perform advanced skills.

Feedback:

- Really values progressive overload tracking (e.g., reps or hold time improvements).
- Feature to consider: Video tracking or photo comparison for form improvements over time.
- Suggests also adding community leaderboard or friend challenge feature for motivation.

Interview 3 - Gym

User background:

Regular gym-goer, tracks weight and reps on paper or in simple logging apps.

Feedback:

- Loves auto-fill from previous sessions
- Missing feature: Workout intensity and volume analysis (e.g., total lifted weight per week).
- Suggests adding trainer comments integrated into session history.